

Inverses from Graphs and Tables

TEACHER KEY | Algebra II | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or a graph, use $f^{-1}(x)$ notation, and recognize that a function and its inverse are reflections across the line $y = x$.

Vocabulary

Inverse of a function — the function that **undoes** the original, written $f^{-1}(x)$. Swap x and y in a table; reflect across $y = x$ on a graph.

Part 1 — Inverses from Ordered Pairs

Swap the coordinates in each pair.

Example 1. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{ (-6, -4), (-1, 0), (4, 2), (-7, 3), (7, 7) \}$

Domain of $f = \{ -4, 0, 2, 3, 7 \}$ which becomes the **range** of f^{-1} .

You Try 1. $g = \{ (1, 5), (2, 8), (3, 11) \}$ $g^{-1} = \{ (5, 1), (8, 2), (11, 3) \}$

Part 2 — Inverses from a Table

Fill in the inverse table by trading the rows.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

Now the inverse:

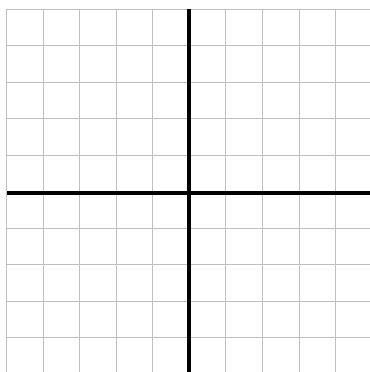
x	-6	-1	4	-7	7
$f^{-1}(x)$	-4	0	2	3	7

The two tables are the same information, just **read the other way**.

Part 3 — Graph a Function and Its Inverse

Each square is 1 unit. Plot f , then plot f^{-1} , then draw the line $y = x$ through the origin at 45° .

A. Plot $f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$ and its inverse.



B. What do you notice?

The two graphs are mirror images across the line **$y = x$**

Which points stay put under the swap?
points already on $y = x$, like (0,0) and (1,1)

If (a, b) is on f , then on f^{-1} you find **(b, a)**

$f^{-1} = \{ (-2, -4), (0, 0), (1, 1), (4, 2) \}$

Part 4 — Domain and Range Trade Places

For any relation: domain of f = **range** of f^{-1} , and range of f = **domain** of f^{-1} .

Function	Domain of f	Range of f	Domain of f^{-1}	Range of f^{-1}
$f(x) = x^3$	all reals	all reals	all reals	all reals
$f(x) = 2^x$	all reals	$(0, \infty)$	$(0, \infty)$	all reals
$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$	$[0, \infty)$	$[0, \infty)$

Row 2 explains Day 7: the exponential and the **logarithm** are inverses, which is why their domains and ranges swapped.

Part 5 — Quick Check

- Does $f^{-1}(x)$ mean 1 divided by $f(x)$? **no**
- If $(3, -5)$ is on the graph of f , name a point on f^{-1} . **$(-5, 3)$**
- A graph and its inverse are reflections across which line? **$y = x$**

Practice

Find each inverse. Show your work where there is room.

- $p = \{ (0, 3), (1, 5), (2, 7) \}$ $p^{-1} = \{ \mathbf{(3, 0), (5, 1), (7, 2)} \}$
- Domain of $p = \mathbf{\{0, 1, 2\}}$ Range of $p^{-1} = \mathbf{\{0, 1, 2\}}$
- A table gives f : $(2, 6), (4, 6), (5, 1)$. Write f^{-1} , then decide whether it is a function.

$f^{-1} = \{ \mathbf{(6, 2), (6, 4), (1, 5)} \}$ A function? **no - 6 has two outputs**

- That is a preview of Day 12. What went wrong in item 3?

- If a graph contains $(-3, 7)$, name the matching point on its inverse. **$(7, -3)$**

Exit Ticket

A function h has the table below.

h : $(1, 4), (2, 9), (5, 0)$ $h^{-1} = \{ \mathbf{(4, 1), (9, 2), (0, 5)} \}$

Domain of h^{-1} : **$\{0, 4, 9\}$** Range of h^{-1} : **$\{1, 2, 5\}$**